



CANDIDATE
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ANSWER KEY

CIVICS
GROUP

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REGISTRATION
NUMBER

H2 Biology

9744/02

Paper 2 Structured Questions

15 September 2021

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

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Total	100

This document consists of **29** printed pages and **0** blank pages.

Answer **all** questions.

- 1** Hormones are responsible for regulating the permeability of kidney cells to water and dissolved ions.

Kidney cells in the collecting duct are impermeable to water. To allow water molecules to move into cells, the hormone vasopressin upregulates the expression of aquaporin channel proteins.

Fig. 1.1 shows the cell surface membrane of a kidney cell, with five proteins labelled **A** to **E**.

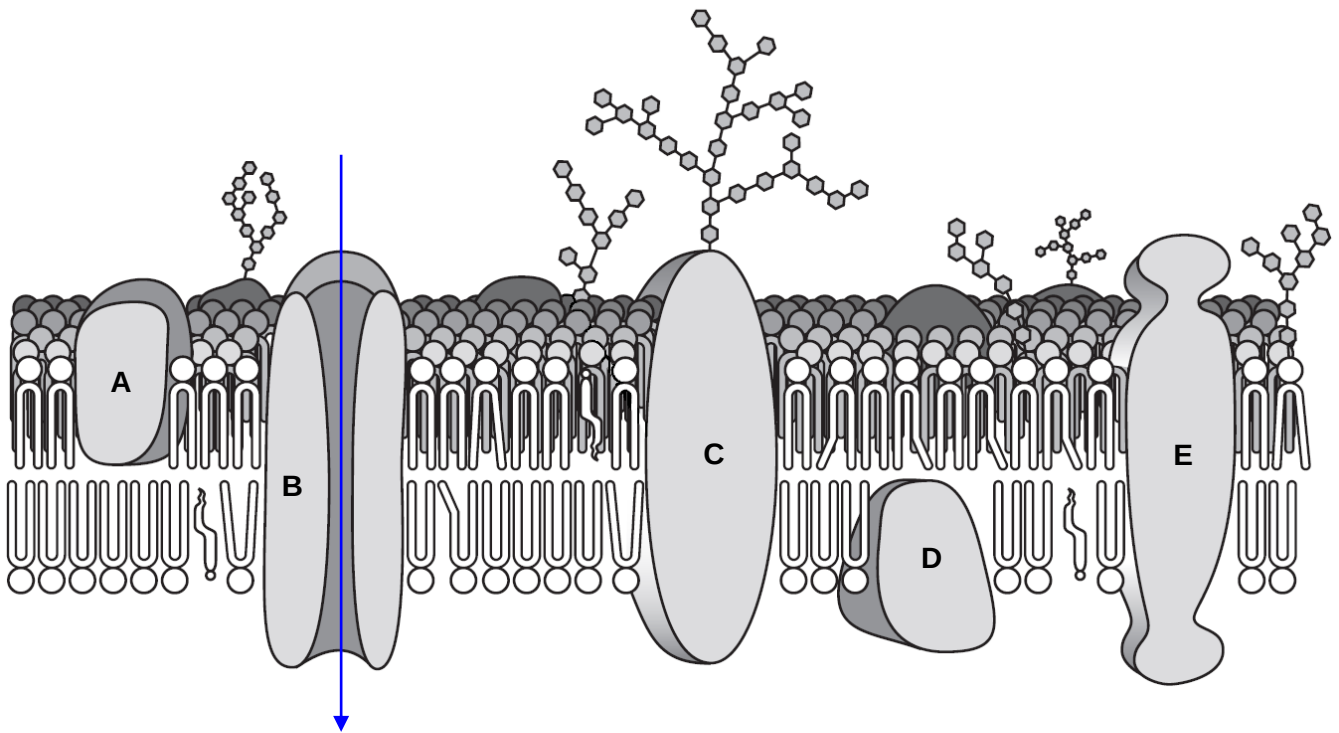


Fig. 1.1

(a) From Fig. 1.1,

- (i)** identify which protein represents the structure of an aquaporin. [1]

B

- (ii)** draw an arrow on Fig. 1.1 to show the direction of movement of water molecules. [1]

Arrow from top to bottom

- (iii)** explain how the arrangement of molecules supports the fluid mosaic model. [2]

1. 'Fluid' = phospholipid and protein molecules are free to move laterally within a layer;
2. 'Mosaic' = random arrangement of the proteins embedded amongst the phospholipid molecules resemble a mosaic pattern;

Vasopressin receptors are G-protein coupled receptors with a transmembrane domain.

(b) Explain how vasopressin receptors are held in the membrane. [2]

1. Phospholipids have a charged and hydrophilic phosphate head which will form hydrogen bonds and ionic bonds with polar and (oppositely) charged amino acids in the protein;
2. The hydrophobic core/fatty acid tails forms hydrophobic interactions with non-polar amino acids in these proteins;

The lipid-soluble hormone aldosterone regulates the level of sodium and potassium ions. It achieves this by upregulating the expression of Na⁺-K⁺ pumps. Fig. 1.2 shows how these protein pumps facilitate the transport of sodium ions into the cell and potassium ions out of the cell.

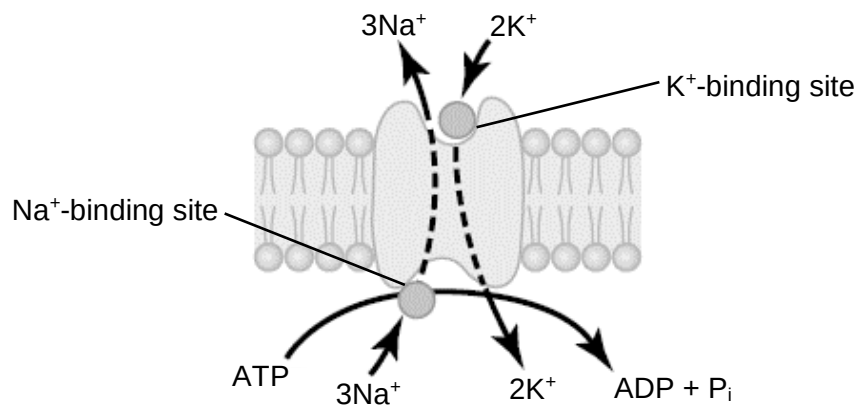


Fig. 1.2

(c) Explain why Na⁺-K⁺ pumps allow Na⁺ and K⁺ to move across the cell surface membrane. [2]

1. Na⁺ and K⁺ ions are charged and hydrophilic, and hence, will be repelled by the hydrophobic core of the cell surface membrane;
 2. Instead, they bind to binding sites on the Na⁺-K⁺ pump that will allow the ions to move from one side to the other;
- A: Na⁺-K⁺ pump utilises energy from ATP hydrolysis to change conformation and pump ions across the cell surface membrane;

(d) Describe **one** difference in the transport of sodium ions and water molecules into kidney cells in response to aldosterone and vasopressin respectively. [1]

Any one:

1. Na⁺ ions are pumped into the cell via **active transport**, while water molecules move into the cell via **osmosis/facilitated diffusion**;
2. Transport of Na⁺ ions into the cell occurs via **Na⁺-K⁺ pumps/protein pumps**, while water molecules diffuse through **aquaporin/channel protein**;
3. Active transport of Na⁺ ions into the cell **utilises energy** released from **ATP hydrolysis** (I: active), while facilitated diffusion of water required **no energy/ATP** (I: passive);
4. Na⁺ ions must bind to the Na⁺-K⁺ pump/protein pump in order to be transported in, while water molecules do not bind to the aquaporin/channel protein;
5. Protein pumps must undergo a change in conformation to release bound Na⁺ ions, while channel proteins do not undergo any conformational change;

[Total: 9]

- 2 Tuberculosis (TB) is an infectious disease caused by the pathogenic bacterium *Mycobacterium tuberculosis*. In this bacterial species, an enzyme known as fructose 1,6-bisphosphate aldolase (FBA) is critical for glucose metabolism.

Fig. 2.1 shows the relative activity of *M. tuberculosis* FBA measured at different pH levels.

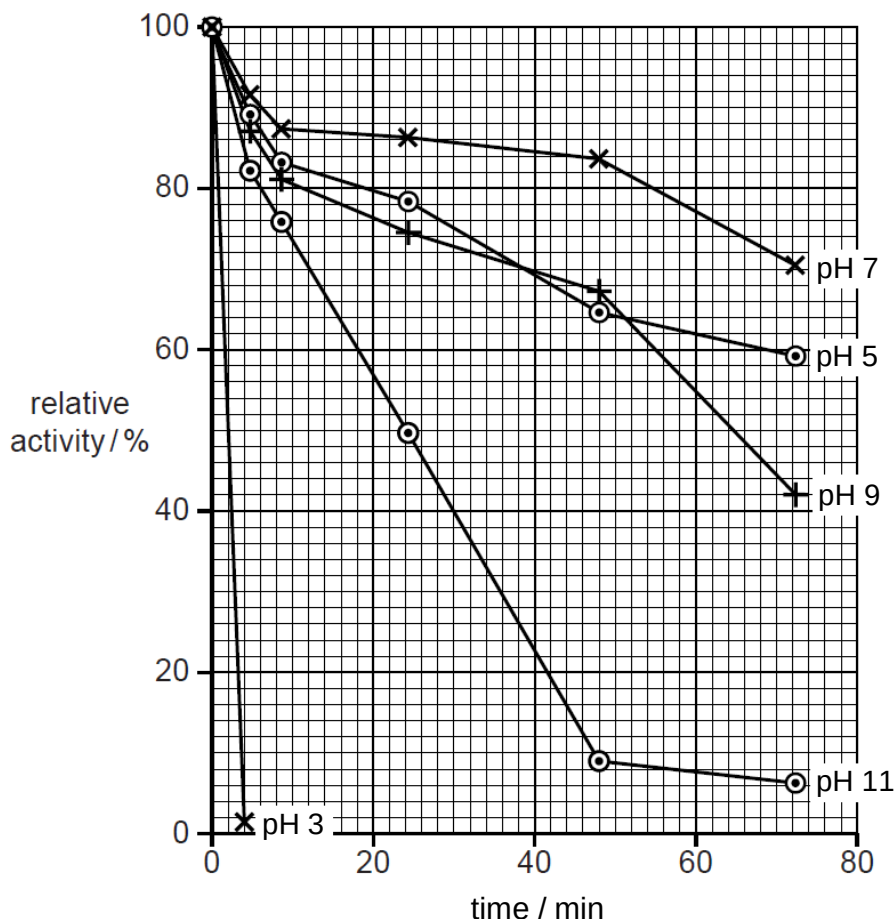


Fig. 2.1

- (a) With reference to Fig. 2.1, distinguish between the effect of neutral and alkaline pH on the activity of FBA. [1]

1. From 0 to 72 min, there was a much greater decline (A: more drastic decline) in relative activity of FBA of 93.5% at pH 11 (A: 58.0% at pH 9) as compared to only 29.5% at pH 7;

OR

2. By 72 min, there was greater FBA relative activity remaining at 70.5% at pH 7 as compared to only 6.0-6.5% at pH 11 (A: 42.0% at pH 9);

Fig. 2.2 shows the structure of FBA, which is made of four identical polypeptide subunits.

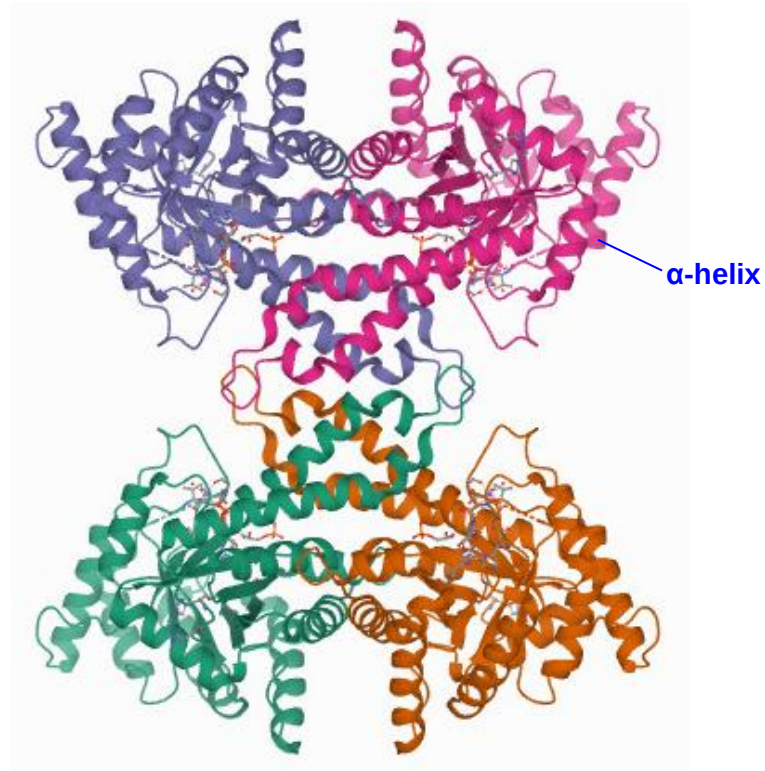


Fig. 2.2

(b) (i) On Fig. 2.2, label and name a secondary structure. [1]

(ii) Explain how changing the pH level affects the structure and activity of FBA. [4]

1. At optimal pH FBA activity is at maximum, and any deviation away from optimal pH would result in sharp decrease in FBA activity;
2. Because at optimal pH, the 3D conformation of FBA's active site is complementary to substrate / fructose 1,6-bisphosphate;
3. However, changes in pH / H^+ concentration can disrupt / break hydrogen and ionic bonds between R groups, hence causing a change in 3D conformation of the active site;
A: excess H^+ and OH^- ions neutralise negatively charged and positively charged amino acid R groups respectively, hence changing the 3D conformation and charge of FBA active site;
4. And hence, FBA is denatured as active site is no longer complementary in conformation and charge to substrate molecules; (R: active site is/bonds are denatured)

Recent research has identified a potential therapeutic drug that targets the FBA enzyme known as 8-hydroxyquinoline carboxylic acid (HCA). Fig. 2.3 shows the molecular shapes of HCA and fructose 1,6-bisphosphate, which is the substrate for FBA.

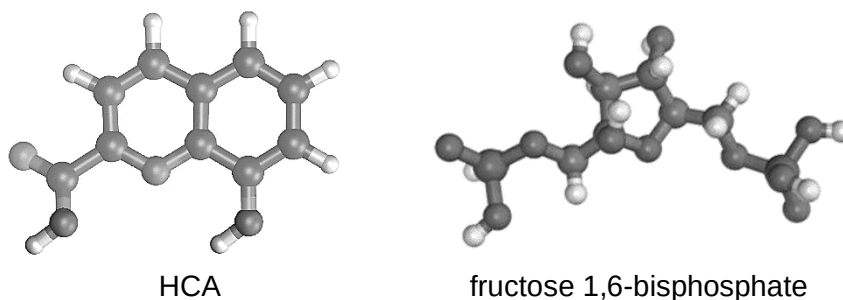


Fig. 2.3

(c) Suggest how HCA may act as a therapeutic drug to treat tuberculosis. [4]

1. HCA acts as a non-competitive inhibitor to FBA;
2. Because its molecular shape / 3D conformation is not similar to fructose 1,6-bisphosphate;
3. Hence, HCA binds to a site other than the active site to result in a change in 3D conformation of FBA active site, thus inhibiting enzyme activity;
4. Resulting in *M. tuberculosis* cells being unable to carry out glucose metabolism, thus killing the bacteria cells;

OR

5. HCA acts as an allosteric inhibitor to FBA;
6. Because its molecular shape / 3D conformation is not similar to fructose 1,6-bisphosphate;
7. Hence, HCA binds to allosteric site to stabilise the inactive conformation of FBA, thus inhibiting enzyme activity;
8. Resulting in *M. tuberculosis* cells being unable to carry out glucose metabolism, thus killing the bacteria cells;

[Total: 10]

3 Fig. 3.1 shows the normal differentiation pathway of a spermatogonial stem cell into spermatids.

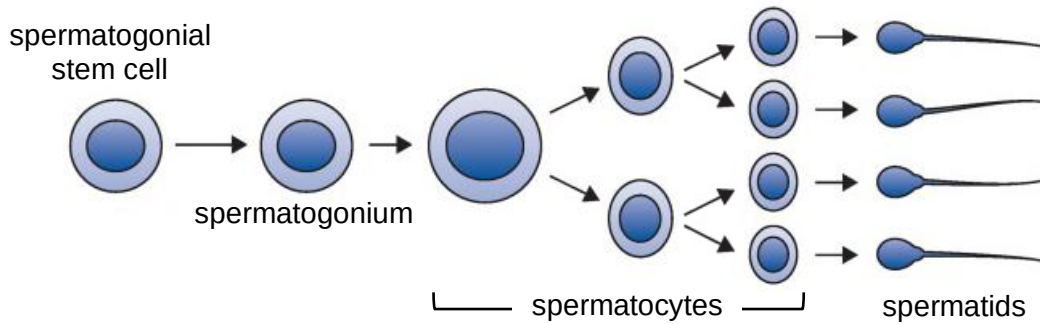


Fig. 3.1

Fig. 3.2 shows three groups of spermatocytes, **A**, **B** and **C**, in different stages of cell division. The peak relative mass of DNA is indicated for each group of cells.

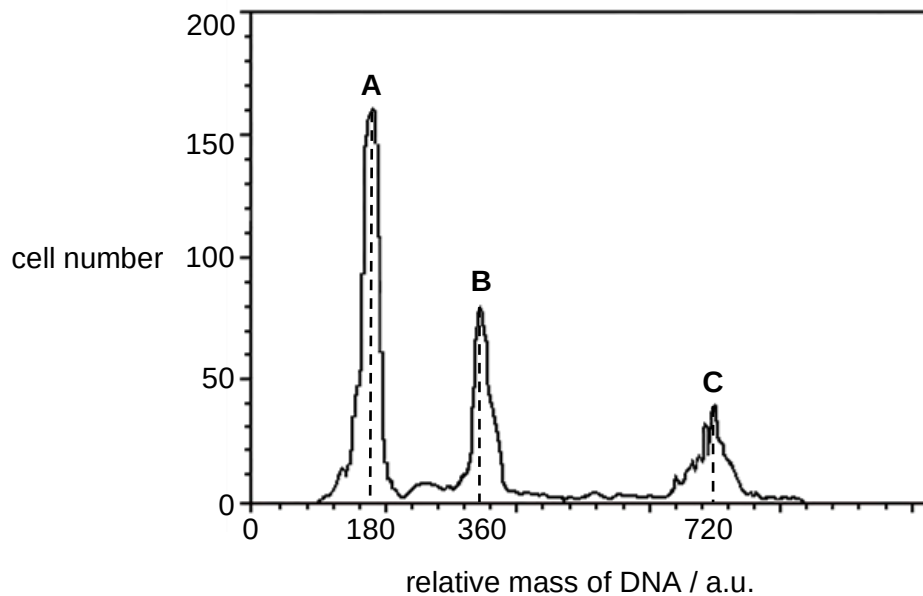


Fig. 3.2

(a) With reference to Fig. 3.2,

(i) identify the group of spermatocytes that are formed at the end of

[1]

meiosis I	meiosis II
B	A

(all or none)

(ii) explain your answer in (a)(i).

[3]

1. Spermatocyte undergoes meiosis I and meiosis II, where relative mass of DNA is halved from 720 to 360 a.u. and halved again from 360 to 180 a.u. respectively;
2. Because pairs of homologous chromosomes are first separated into two daughter cells at the end of meiosis I;
3. And subsequently, sister chromatids (which are now daughter chromosomes) are separated into two gametes/spermatids at the end of meiosis II;

Germ cell tumours result from uncontrolled division of germ cells, which give rise to gametes. In males between the ages of 15 to 35 years, the risk of testicular cancer increases with age.

Fig. 3.3 shows how a tumour may result from uncontrolled cell division of a spermatogonium (male germ cell), instead of normal differentiation into spermatids.

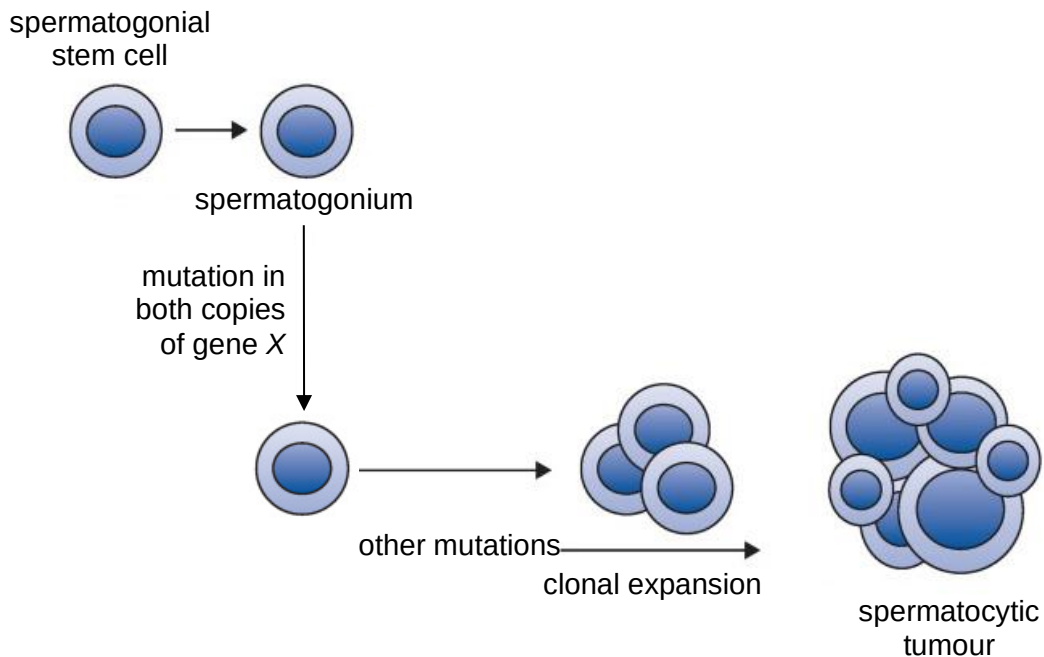


Fig. 3.3

(b) (i) Suggest the name of gene X. [1]
p53

(ii) Explain how gene X may result in uncontrolled cell division of a spermatogonium. [3]

1. A loss-of-function mutation of both alleles of p53 / tumour suppressor gene,
 2. Which results in failure to arrest cell cycle / activate DNA repair mechanisms;
 3. Thus leading to accumulation of (other) mutations in a single cell, which leads to dysregulation of cell cycle checkpoints;
- A: gain-of-function mutation in proto-oncogenes and other tumour suppressor genes leads to dysregulation of cell cycle checkpoints;

(c) Cancer treatments, including chemotherapy and radiotherapy, can often result in loss of fertility. In male cancer patients, sperm cryopreservation (freezing of sperm cells) is one method of preserving fertility.

Using your knowledge of stem cells, suggest why fully differentiated sperm cells are preserved instead of spermatogonial stem cells. [1]

Any one:

1. Spermatogonial stem cells are much fewer in number than sperm cells, therefore harder to extract;
2. Spermatogonial stem cells must be induced using appropriate molecular signals to undergo differentiation into specialised sperm cells, which requires an additional step;
3. Any existing mutations within a spermatogonial stem cell will be passed on to all sperm cells that result from meiotic division;
4. AVP;

[Total: 9]

- 4 In the bacterium *Thermus thermophilus*, a sequence of DNA contains two overlapping genes, *rnpA* and *rpmH*, which code for RNase P and a ribosomal protein respectively.

Fig. 4.1 shows the first 30 bases in the DNA sequence. Following process X, a different strand of nucleic acid is formed, and it contains the translation start sites for RNase P and the ribosomal protein.

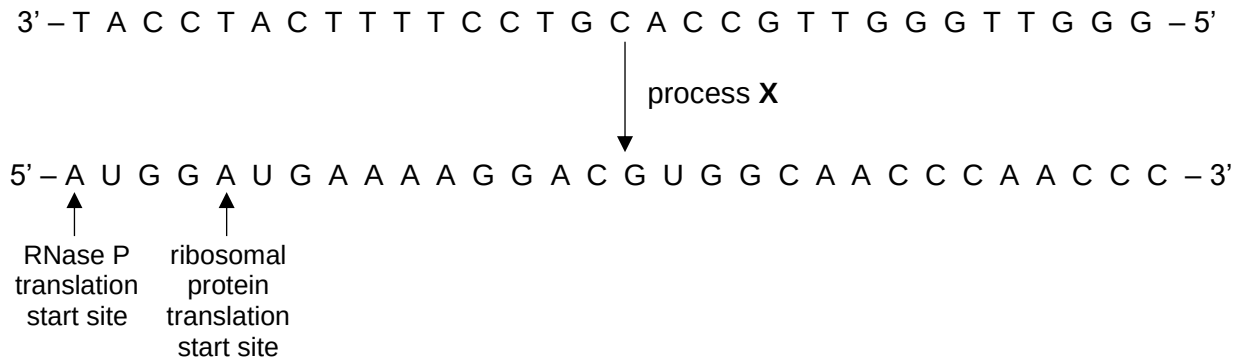


Fig. 4.1

- (a) (i) Briefly describe process X. [3]

1. RNA polymerase binds to the promoter to initiate transcription;
2. It unzips double stranded DNA, and uses one DNA strand as a template to synthesise an mRNA strand;
3. Via complementary base pairing between adenine & uracil, cytosine & guanine, thymine & adenine (must have all three);
4. It catalyses the formation of phosphodiester bonds between ribonucleotides to form the mRNA strand;
5. mRNA strand is synthesised in the 5' to 3' direction (A: template DNA strand is read in the 3' to 5' direction);

- (ii) Suggest why the product of process X can code for two different polypeptide products. [2]

1. After transcription of this DNA sequence, there are two start codons AUG in the mRNA strand;
2. Ribosome begins translation four nucleotides apart / in different reading frames / reads a different sequence of mRNA codons (OWTTE), hence producing two different amino acid sequences can be produced from the same mRNA strand;

In eukaryotic cells, the *FECH* gene codes for ferrochetalase, which is the final enzyme in the metabolic pathway of haem synthesis.

A study done in zebrafish found that a point mutation in the *FECH* gene resulted in a disease phenotype known as dracula. The disease is characterised by lysis of red blood cells upon exposure to light.

Fig. 4.2 shows a segment of the *FECH* template DNA sequence in normal and dracula zebrafish, and Table 4.1 shows the genetic code.

normal

27	28	29	30	31	32	33
CAC	GTG	GAC	<u>AGT</u>	GGA	CTC	ATC

dracula

27	28	29	30	31	32	33
CAC	GTG	GAC	<u>ATT</u>	GGA	CTC	ATC

Fig. 4.2

Table 4.1

	U		C		A		G		
U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys	U
	UUC		UCC		UAC		UGC		C
	UUA	Leu	UCA		UAA	STOP	UGA	STOP	A
	UUG		UCG		UAG	STOP	UGG	Trp	G
C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg	U
	CUC		CCC		CAC		CGC		C
	CUA		CCA		CAA	Gln	CGA		A
	CUG		CCG		CAG		CGG		G
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser	U
	AUC		ACC		AAC		AGC		C
	AUA	Met	ACA		AAA	Lys	AGA	Arg	A
	AUG		ACG		AAG		AGG		G
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly	U
	GUC		GCC		GAC		GGC		C
	GUA		GCA		GAA	Glu	GGA		A
	GUG		GCG		GAG		GGG		G

(b) Using Fig. 4.2 and Table 4.1,

(i) State the amino acid sequence for this segment of the normal polypeptide. [1]
(N-terminus) **Val – His – Leu – Ser – Pro – Glu** (C-terminus)

(ii) Explain how the point mutation results in non-functional ferrochetalase. [2]

1. The point substitution mutation, where G is replaced with T, introduces a **premature stop codon UAA** (instead of UCA - Serine), thus leading to a nonsense mutation;
2. Translation is prematurely terminated, resulting in a **truncated polypeptide**, which then folds into a protein of **different 3D conformation** that is non-functional;

[Total: 8]

- 5 Hantaviruses are a group of viruses that normally infect rodents, but do not cause disease in them.

Humans may become infected with Hantaviruses through contact with rodent urine, saliva, or faeces. Some strains cause potentially fatal diseases, such as Hantavirus Haemorrhagic Fever with Renal Syndrome (HFRS), and Hantavirus Cardiopulmonary Syndrome (HCPS), in humans.

Fig. 5.1 shows the structure of a strain of Hantavirus. The genome of the virus consists of three segments – small, medium and large.

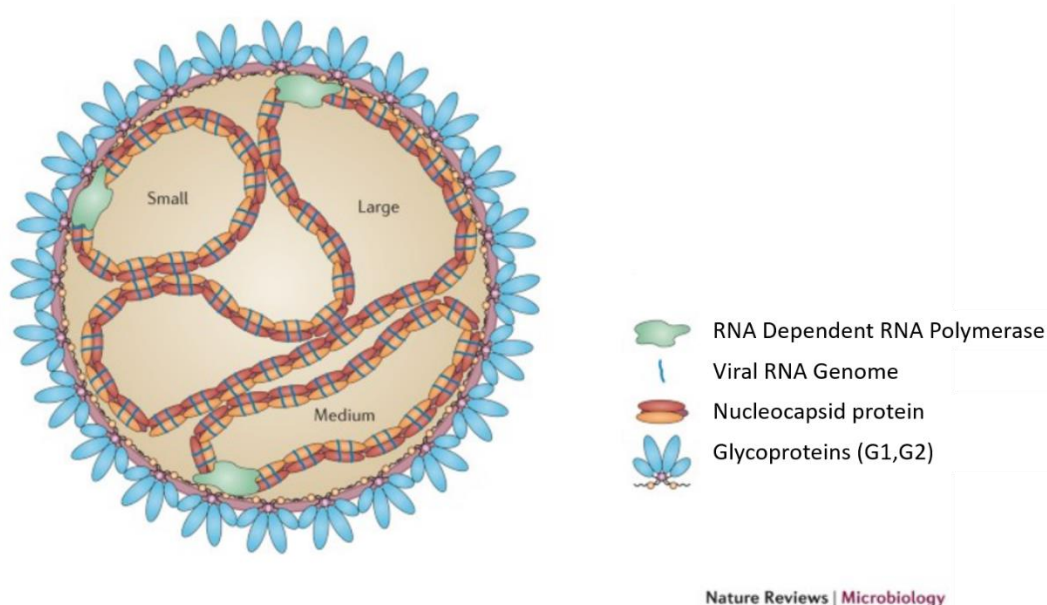


Fig. 5.1

- (a) (i) With reference to Fig. 5.1, describe **three** structural features which distinguish the Hantavirus from its human host cell. [3]

Feature	Hantavirus	Human Host Cell
Genome	Made up of <u>ribonucleotides</u> (RNA)	Made up of <u>deoxyribonucleotides</u> (DNA)
Enzyme for genome replication OR Enzyme for mRNA synthesis	Has <u>RNA-dependent RNA polymerase</u> for viral RNA genome replication Has <u>RNA-dependent RNA polymerase</u> for viral mRNA synthesis	Uses <u>DNA polymerase</u> for DNA replication <u>DNA-dependent RNA polymerase</u> for transcription
Structure of genome	RNA segments/genome is <u>circular/joined at the ends</u>	Chromosomes/genome is <u>linear</u>
Genome organisation	RNA genome consists of <u>three segments</u>	Genome made up of <u>46 chromosomes</u>
Proteins bound to genome	<u>Nucleocapsid protein</u> bound to viral RNA genome	<u>Histone proteins</u> bound to human DNA genome

AVP:

Hantavirus replication takes place in macrophages and vascular endothelial cells, especially in the lung and the kidney. Fig. 5.2 shows the reproductive cycle of the Hantavirus.

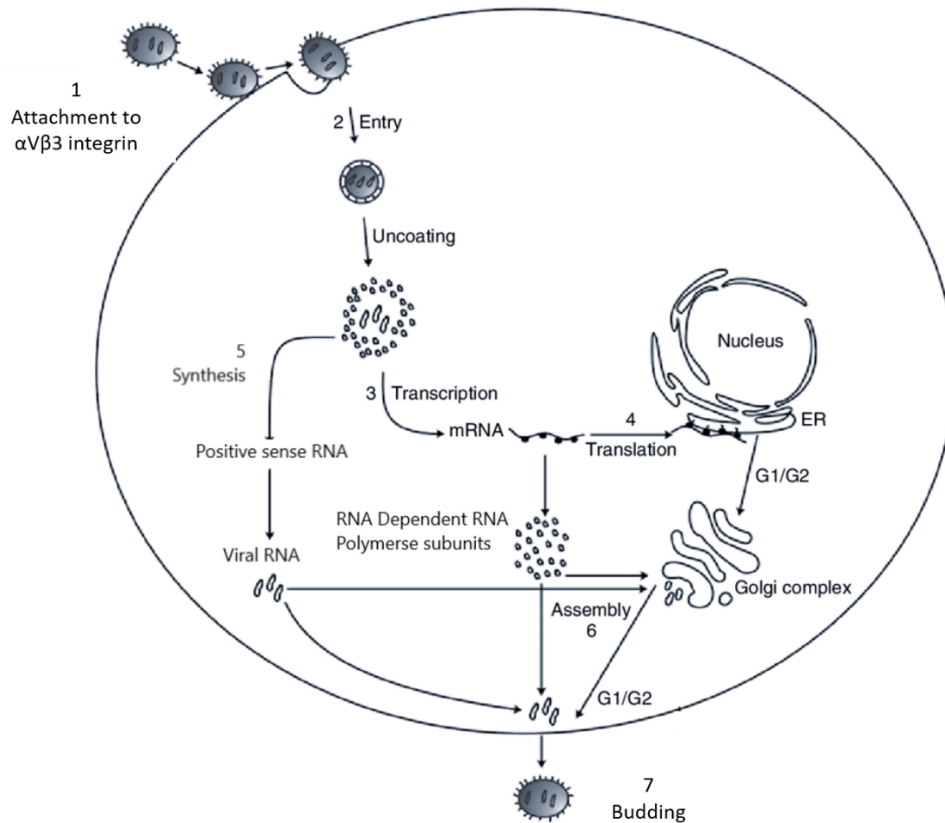


Fig. 5.2

(ii) With reference to Fig. 5.2, name another virus which has a similar reproductive cycle to the Hantavirus. [1]

1. Influenza virus
2. AVP: named negative sense ssRNA virus with RNA dependent RNA polymerase e.g. Rabies, etc.

(iii) Describe how the reproduction of the Hantavirus is similar to the virus named in (a)(ii). [3]

1. Attachment – both require surface **glycoproteins** to attach to specific host cell receptors;
R: haemagglutinin in Influenza attaches to sialic acid receptors whereas G1/G2 proteins attach to $\alpha V\beta 3$ integrin proteins in Hantavirus (phrased as a difference)
2. (Replication) Synthesis of viral mRNA – viral RNA-dependent RNA polymerase is used for synthesis of viral mRNA (positive sense) using the viral RNA genome as a template;
3. (Replication) Replication of viral RNA genome – viral-RNA dependent RNA polymerase is used for synthesis of more copies of the viral genome using positive sense ssRNA/mRNA as a template;
4. (Replication) Protein synthesis – both viruses utilise host cell ribosomes at the rough endoplasmic reticulum for the synthesis of viral proteins;
5. (Assembly) Embedding – both viruses have their surface glycoproteins embedded on the host cell surface membrane;

R: influenza has haemagglutinin and neuraminidase embedded whereas G1 & G2 proteins are embedded for Hantavirus (phrased as a difference)

6. (Release) Exit – both viruses exits/released from host cell via budding
7. (Release) Exit – both viruses acquire their viral envelope from host cell surface membrane with embedded viral glycoproteins;
8. AVP: glycosylation of viral proteins to form glycoproteins at Golgi body

- (b) Human Hantavirus infections are assumed to occur accidentally, and humans represent a dead end host for the Hantavirus. Transfer of virus particles from infected to uninfected people normally does not occur.

One exception is the Andes Hantavirus (ANDV) strain found in South America, for which sporadic person-to-person transmissions have been reported. This finding reveals a worrying risk potential of Hantaviruses for human health.

- (i) Describe how the ANDV strain becomes capable of human-to-human transmission. [2]

1. Antigenic drift;
2. The ANDV strain accumulated mutations over time in its genome and this led to changes in the 3D conformation and charge of the viral proteins which eventually allowed for effective human-to-human transmission

Fig. 5.3 shows the data for ANDV infections, as reported by the Chilean Ministry of Health. The solid line represents the effective number of ANDV infections in the population. The bar graph represents the total number of human Hantavirus Cardiopulmonary Syndrome (HCPS) cases per year.

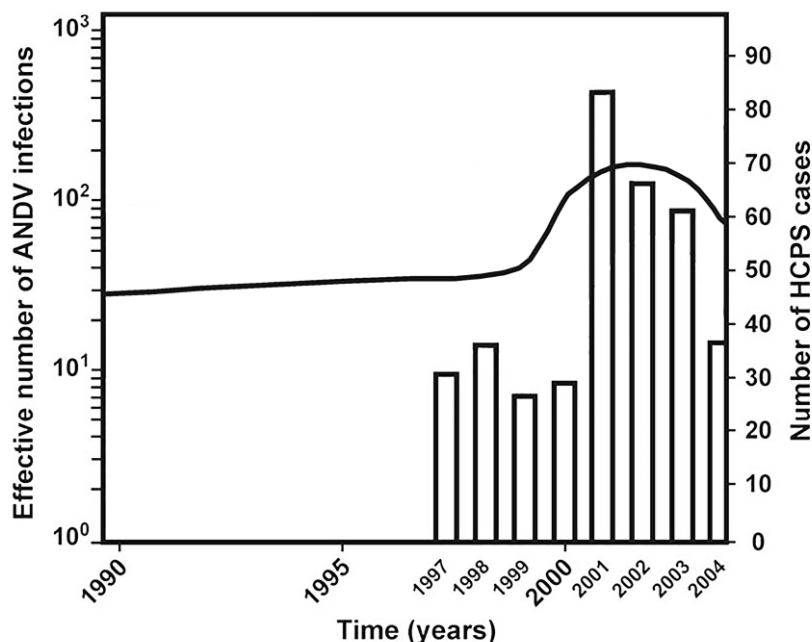


Fig. 5.3

- (ii) Suggest why there are no human HCPS cases reported before the year 1997. [1]

1. The ANDV strain was not capable of causing HCPS in humans before 1997;
2. Data collection for HCPS started only in 1997, and thus, earlier reported data was not available;
3. AVP

[Total: 10]

6 Fig. 6.1 illustrates the *lac* operon and its regulatory gene in *E. coli*.

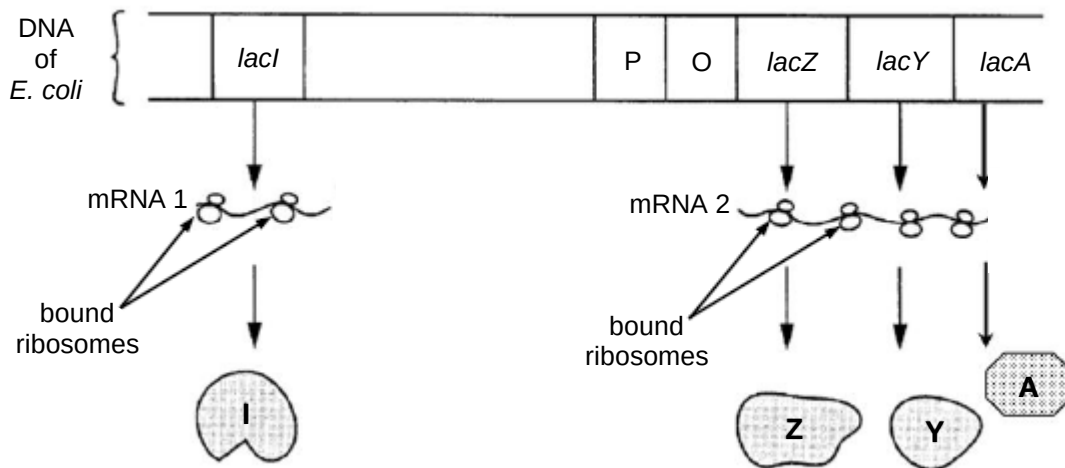


Fig. 6.1

(a) With reference to Fig. 6.1, explain why ribosomes bound to mRNA 1 produce the same polypeptide but those bound to mRNA 2 produce three different polypeptides. [2]

1. mRNA 1 has one start codon and one stop codon / monocistronic and hence, it is translated to form/encodes for the same polypeptide;
2. mRNA 2 contains three start codons and three stop codons (i: polycistronic), and hence, three different polypeptides can be translated from / encoded by one mRNA;

(b) A certain bacteriophage is known to insert its own genome within the *lacZ* gene of *E. coli*. In an experiment, *E. coli* cells were infected with the bacteriophages and the infected cells grown in a medium containing equal concentrations of glucose and lactose.

Fig. 6.2 shows how the number of infected *E. coli* cells varies with concentration of glucose and lactose.

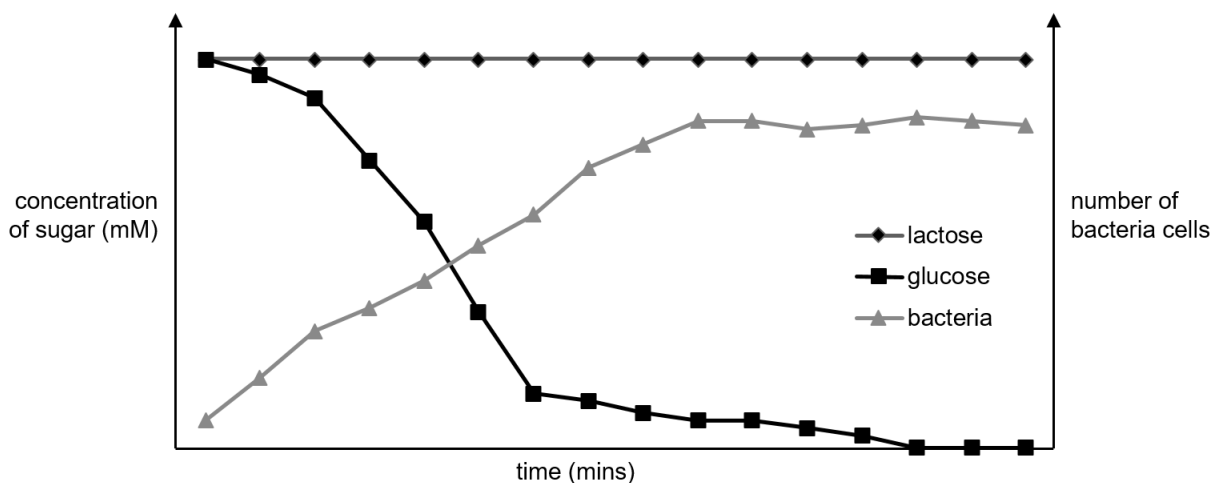


Fig. 6.2

With reference to Fig. 6.2, explain the trend observed in each of the three graphs. [3]

1. increase in number of *E. coli* cells as bacteria reproduce by binary fission;
2. decrease in glucose concentration as it is preferred respiratory substrate used by bacteria;

3. no change in concentration of lactose as *lac* operon not expressed, and hence no β -galactosidase produced;

- (c) The *xyl* operon contains genes involved in the breakdown of the sugar xylose. Fig. 6.3 shows the organisation of the *xyl* operon in another species of bacterium, *T. halophila*.

The *xyl* operon contains three structural genes (*xylA*, *xylB* and *xylE*) and is regulated by the regulatory gene, *xylR*. The arrows represent the direction of transcription of the named genes.

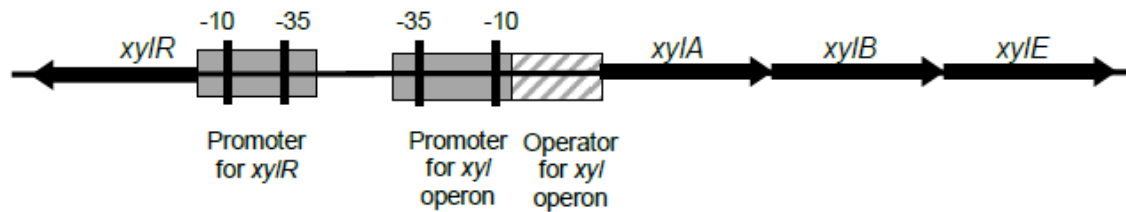


Fig. 6.3

- (i) State whether the *xyl* operon is an inducible or repressible operon. [1]
Inducible operon
- (ii) Suggest how the *xylR* gene regulates the expression of the *xyl* operon when xylose is present. [4]
- xylR* gene codes for *xyl* repressor protein;
 - xylose binds to allosteric site of *xyl* repressor protein causing it to undergo conformational change to become inactive;
 - Repressor protein is unable to bind to operator (R: promoter) of *xyl* operon, and hence, RNA polymerase can bind to promoter;
 - Hence, structural genes are expressed / transcription of *xylA*, *xylB* and *xylE* can proceed, and xylose-catabolising enzymes are only synthesised in the presence of xylose;

[Total: 10]

- 7 Chromosome 5-related Spinal Muscular Atrophy (SMA) is caused by a mutation in the Survival Motor Neuron 1 (SMN1) gene. This gene is responsible for producing the Survival Motor Neuron (SMN) protein, which maintains the health and normal function of motor neurons.

In individuals with SMA, both copies of the SMN1 gene are mutated, leading to decreased production of SMN protein. Without proper levels of SMN protein, motor neurons in the spinal cord will be lost, preventing the muscles from receiving proper signals from the brain as shown in Fig. 7.1.

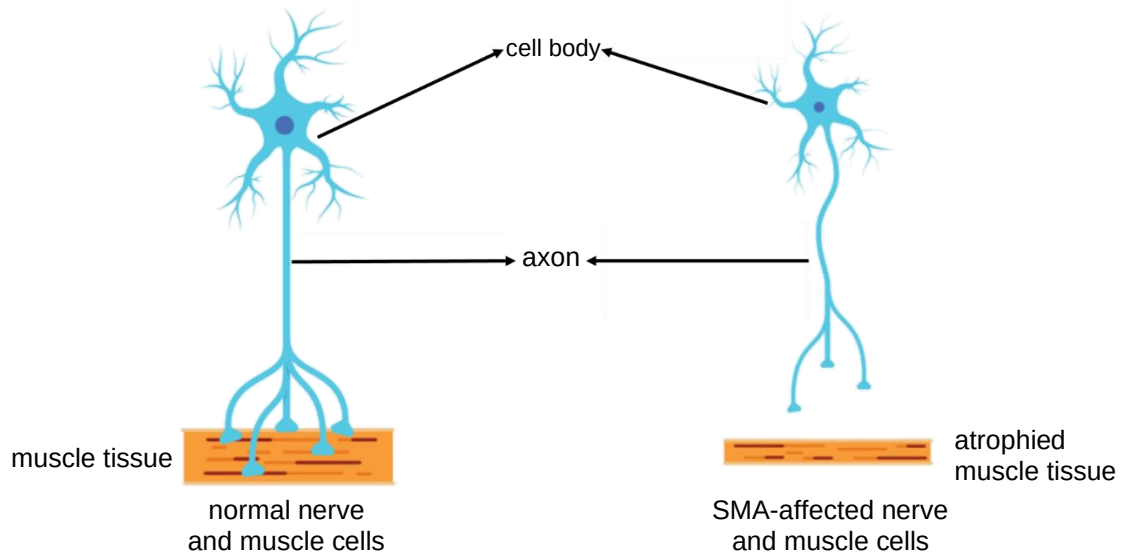


Fig. 7.1

- (a) Explain if the SMN1 mutation resulting in SMA is dominant or recessive in nature. [2]
1. Recessive mutation;
 2. Both copies of SMN1 gene need to be mutated for the SMA phenotype to be exhibited

- (b) A neighbouring gene on chromosome 5, called SMN2, also produces SMN protein. Most of the protein expressed by SMN2 genes is not functional, but a small percentage, around 15% are functional as illustrated by Fig. 7.2.

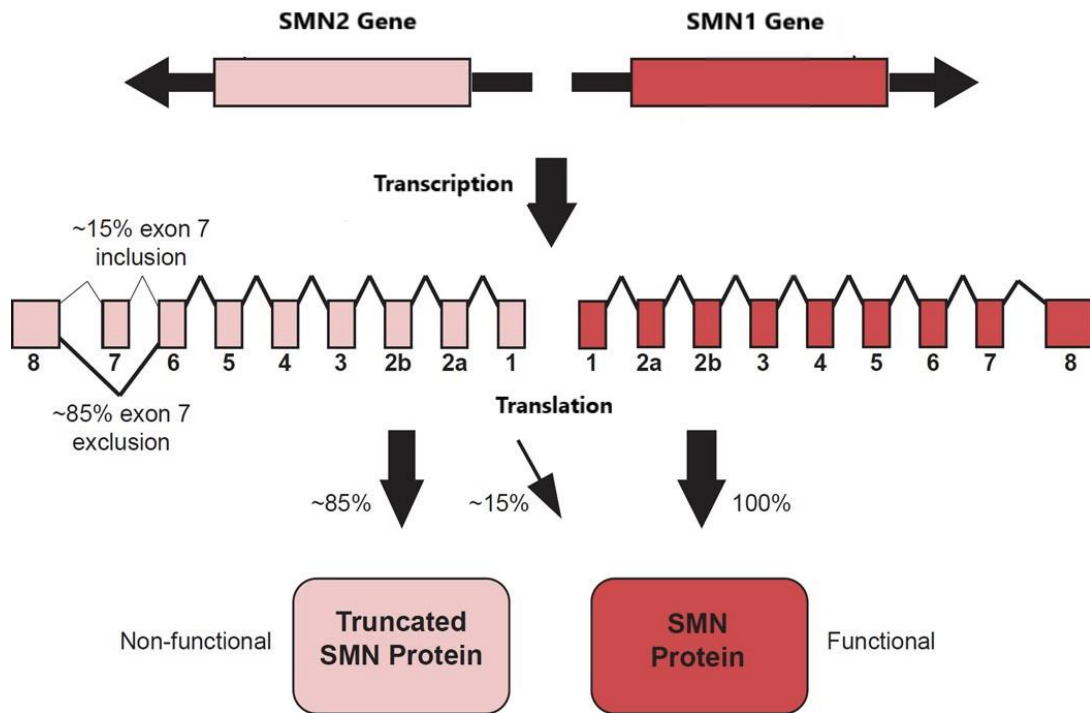


Fig. 7.2

- (i) With reference to Fig. 7.2, explain why 85% of the SMN proteins expressed from the SMN2 gene are synthesised as non-functional in nature. [3]

1. Alternative splicing of SMN2 mRNA;
2. 85% of the pre-mRNA are spliced to have exon 7 removed in the pre-mRNA and have lesser codons than the full length mature mRNA;
3. This results in a truncated polypeptide with fewer amino acids present;

Individuals can have multiple copies of the SMN2 gene. Normally, the number varies between zero and eight copies.

In the chromosome 5-related form of SMA, having three or more copies of the SMN2 gene is associated with a less severe disease manifestation.

- (ii) State the type of genomic modification which results in multiple copies of SMN2 gene in an individual. [1]

Gene amplification

- (iii) Suggest how having higher copies of SMN2 gene results in less severe disease manifestation. [1]

Higher copy number leads to more functional SMN proteins being expressed by the SMN2 gene. More functional SMN proteins are able maintain the health and normal function of motor neurons hence reducing the severity of the disease.

- (c) Telomeres are found at the end of eukaryotic chromosomes.

Telomeres consist of a nucleotide sequence that is usually repeated about 3,000 times. Telomere-binding proteins are also found to be attached to these telomeric sequences.

Fig. 7.3 shows part of a telomere found in eukaryotic chromosomes.

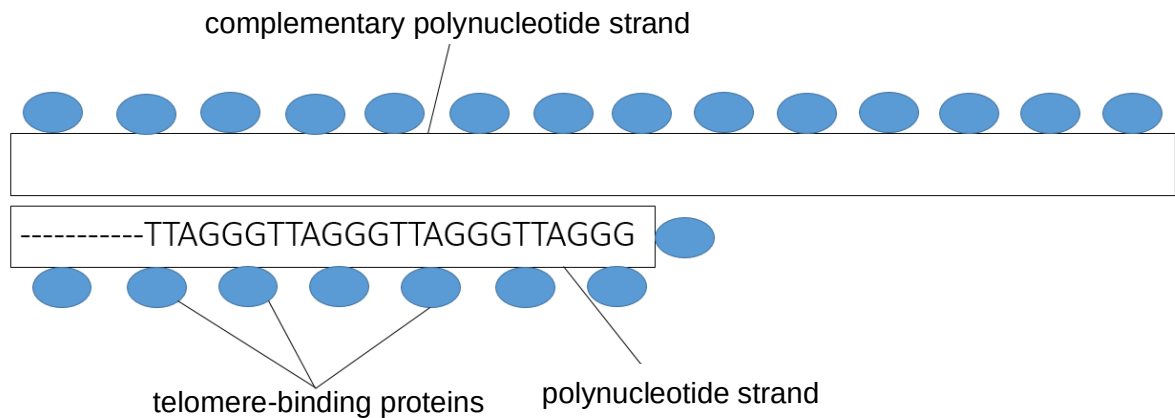


Fig. 7.3

- (i) With reference to Fig. 7.3, state the nucleotide sequence which is repeated in the complementary polynucleotide strand. [1]

5' – AATCCC – 3'

- (ii) Suggest the role of telomere-binding proteins. [1]

1. Protect the DNA from degradation by nucleases;

- (iii) In the past, non-coding sequences were referred to as “junk DNA” which have no purpose in the genome.

Explain why the term “junk DNA” is misleading in the context of the telomere. [2]

Despite being non-coding sequences, telomeres serve important functions:

1. Telomeres ensure genes / vital genetic information are not lost/eroded with each round of DNA replication due to the end replication problem;
2. Telomeres protect and stabilise the terminal ends of chromosomes and prevent DNA damage / formation of telomeric loop prevents fusion of chromosomes;
3. Shortening of telomeres also indirectly prevents the development of cancer because cells may undergo apoptosis (programmed cell death) when their telomeres have shortened to a critical length;

[Total: 11]

- 8 In the presence of light, potatoes produce a greening response. This results in the potato producing broad green leaves, short sturdy stems and long roots as shown in Fig. 8.1.



Fig. 8.1

Upon reception of light by the phytochrome receptor, two signal transduction pathways are elicited in a potato cell as shown in Fig. 8.2. Both pathways eventually result in the synthesis of greening response proteins such as photosynthetic enzymes as well as enzymes involved in chlorophyll production.

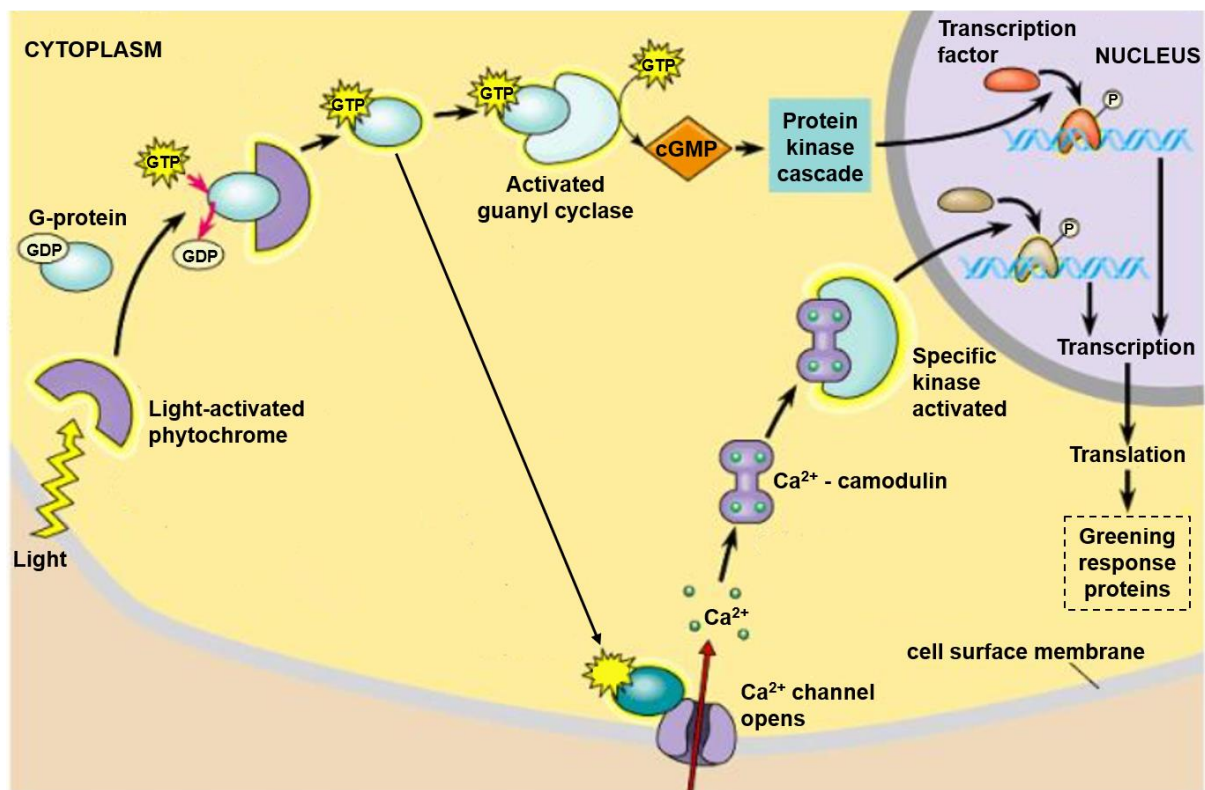


Fig. 8.2

- (a) Describe the role of second messengers.

[2]

1. Second messenger binds and induces a conformation change in the protein kinase / relay protein and the protein kinase is activated;
2. In this way, the 'signal' is passed/relayed to the protein kinase/relay protein during signal transduction;

(b) With reference to Fig. 8.2,

(i) identify the second messengers involved in the signalling pathway. [1]

cGMP and Ca^{2+} / calcium ions; (all or none)

(ii) explain **one** step of the signal transduction stage in the cytoplasm that illustrates signal amplification. [2]

1. The conversion of GTP to cGMP by guanyl cyclase / activation of G protein by phytochrome / protein kinase cascade;

2. A small number of activated guanyl cyclase can result in the production of a large amount of cGMP;

OR

A small number of activated phytochrome can result in the activation of a large number of G-proteins;

OR

Each protein kinase can result in activation of a large number of downstream protein kinases;

(iii) compare the activation of calmodulin with that of the final protein kinase in a protein kinase cascade. [3]

Similarity:

1. Both calmodulin and the protein kinase undergo conformational change to become activated;

Differences:

2. Calmodulin is activated through the binding of 4 calcium ions;

3. Protein kinase is activated through phosphorylation by an upstream protein kinase;

Phytochromes are able to detect red light and far-red light.

An experiment was conducted to investigate the role of phytochromes and germination of lettuce seeds. The lettuce seeds were exposed to alternating periods of red light and far-red light for 1 minute each.

The results of the investigation are shown in Fig. 8.3.

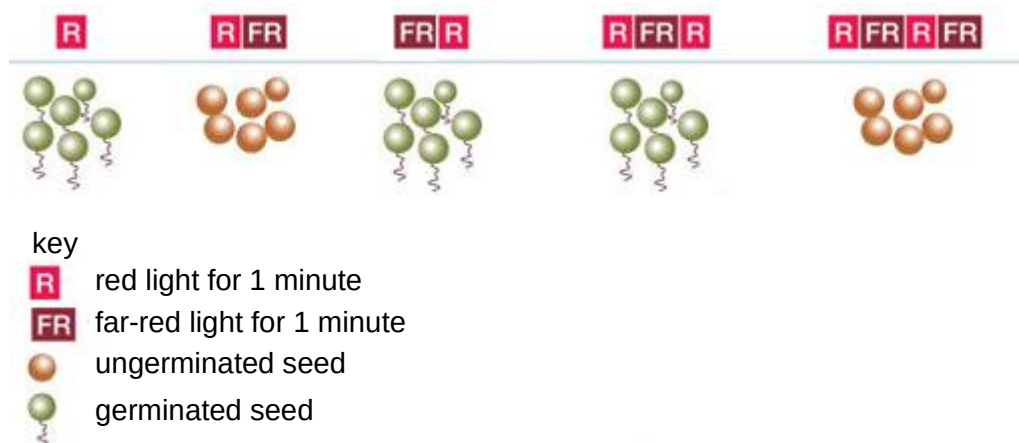


Fig. 8.3

(c) Explain which light, when detected by phytochromes, is more effective in inducing seed germination. [2]

1. Red light is more effective than far-red light; (R: far red is **less** effective – NATQ)
2. When the last light detected by the seeds is red light, the seeds germinate;
(A: When the last light detected by the seeds is far red light, the seeds do not germinate)

[Total: 10]

- 9 Acute Immune Deficiency Syndrome can be caused by either of two retroviruses, known as Human Immunodeficiency Virus types 1 and 2 (HIV-1 and HIV-2).

Related viruses isolated from a number of other primate species are termed as Simian Immunodeficiency Virus (SIV).

Fig. 9.1 shows a phylogenetic tree illustrating the evolutionary relationships among the two major forms of HIV, HIV-1 and HIV-2, and SIV that afflicts non-human primates.

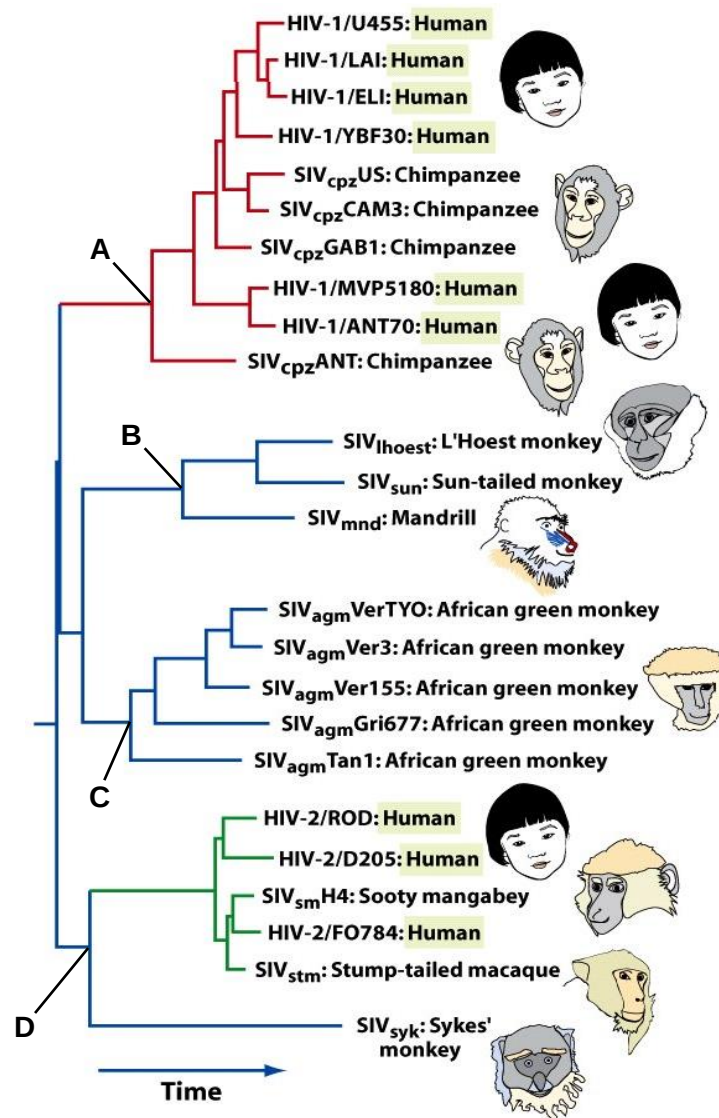


Fig. 9.1

- (a) Using the information given in Fig. 9.1,

- (i) briefly describe how the phylogenetic tree was constructed. [2]

1. Involves a comparison of homologous DNA sequences / amino acid sequences of SIV and HIV strains (multiple sequence alignment);
2. Viral strains that share a greater degree of similarity are more closely related, and hence, grouped more closely together;

- (ii) Nodes A, B, C and D have been labelled on Fig. 9.1 Using the information given, identify the **two** nodes which denote the common ancestors of HIV. [1]

A & D (all or none)

- (b) Fig 9.2 shows a molecular process occurring within an HIV-infected host cell. An anti-viral drug AZT triphosphate was introduced into the cell and the process below was observed.

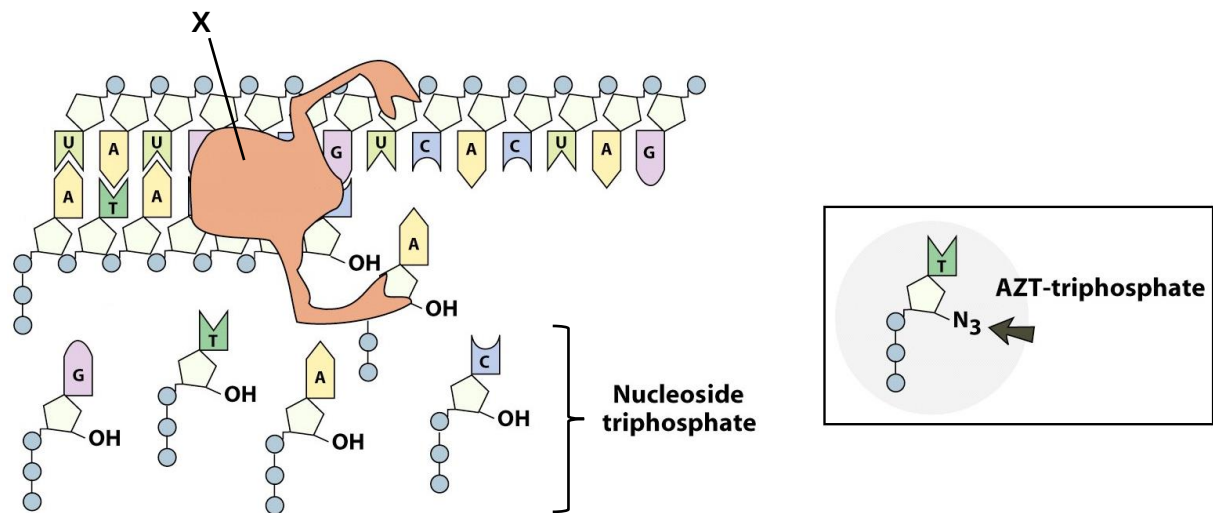


Fig. 9.2

- (i) Identify molecule X and briefly describe the reaction it is involved in as seen in Fig. 9.2 [2]
1. **Reverse transcriptase** (must have);
 2. Uses a single strand of RNA as a template to synthesize a complementary DNA sequence via complementary base pairing;
- (ii) Explain how AZT-triphosphate serves as an anti-viral drug. [2]
1. AZT-triphosphate is similar in 3D-conformation to dTTP / is complementary in 3D-conformation to the active site of the reverse transcriptase / contains a thymine base;
 2. AZT-triphosphate lacks a 3' OH group, and hence, when it is added to the growing DNA strand will stop strand elongation

- (c) Table 9.1 shows the percentage of AZT resistant viral strains in a group of patients who underwent AZT therapy. The percentage of AZT resistant viral strains was assessed at 1, 2, 12 and 18 months after the start of AZT therapy.

Table 9.1

time after start of therapy / months	AZT resistant viral strains / %
1	2
2	10
12	20
18	50

With reference to Table 9.1, explain how the use of AZT therapy can lead to the evolution of AZT resistant viral strains. [4]

1. [Description of data]: With increasing time in therapy, the percentage of AZT resistant strains in the patients increases from 2% in month 1 to 50% in month 18 (need to quote data).
2. Mutations in HIV led to AZT resistance in the population hence leading to variation;
3. AZT acts as selection pressure and AZT-resistant strains are selected for;
4. HIV strains which are AZT-resistant survive and reproduce at a higher rate and pass on the favourable alleles to newly formed viral progeny;
5. Over time, the frequencies of alleles which confer resistance to AZT increase, leading to increase in the percentage of resistant strains as seen in Table 9.1.

[Total: 11]

- 10** Malaria is a serious disease, thought to affect more than 220 million people around the world. Every year, over three quarters of a million people die from malaria, 90% of whom are in Africa.

Malaria is caused by the malarial parasite which infects red blood cells in the body, as shown in Fig. 10.1.

red blood cell
infected with
malarial parasite

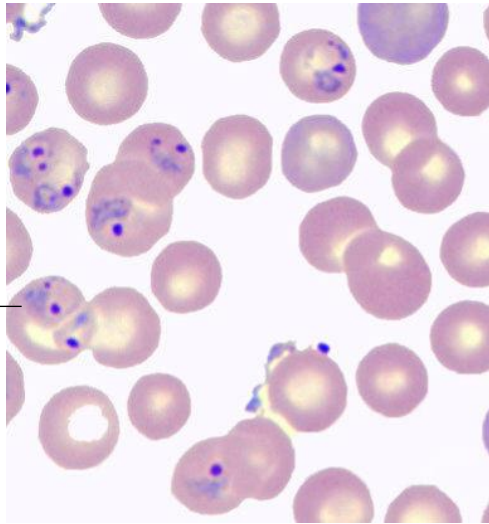


Fig. 10.1

10 to 15 days upon initial infection with the malaria parasite, several symptoms including headache, fever and vomiting will manifest. The disease can rapidly affect many vital organs and become life threatening.

- (a)** After repeated exposure to a variety of malarial strains, some people develop a natural immunity to the disease.

Explain how a person could develop natural immunity to malaria.

[4]

1. Through exposure to a variety of malarial strains, the person will be exposed to antigens from a variety of malarial strains;
2. Specific naïve B / T cells will be activated
(A: Naïve B / T cells with B cell receptor / T cell receptor complementary in conformation to the malarial antigens will be activated)
AND
memory B / T cells will be produced;
3. Upon subsequent infection with malaria parasite in the future, the memory B / T cells will be activated to produce a secondary immune response;
4. A large amount of antibodies against the malarial parasite will be produced quickly to neutralise the malarial parasite;
OR
Large number of cytotoxic T cells will be produced quickly to kill the infected red blood cells, preventing the malarial parasites from replicating in the body;

One way of producing immunity to malaria is through vaccination.

Randomised trials have taken place for a new vaccine called RTS,S. In 2011, 447 children aged 5 to 17 months from Kenya and Tanzania were vaccinated with RTS,S. The children were then monitored to see if they developed malaria. The results are shown in Table 10.1.

Table 10.1

vaccination with RTS,S		
number of subjects	number that developed malaria	percentage that developed malaria
447	38	8.5

The drug company that developed the RTS,S vaccine came to the conclusion that the vaccine had an effectiveness of 56% and suggested that it could be used as an effective vaccination against malaria in the future. Other researchers have suggested that alternative approaches to preventing malaria should be considered.

- (b)** Suggest a reason for and against the introduction of a large-scale vaccination programme using RTS,S.

For

[1]

1. Less people will get the malaria disease / less will die from malaria;
2. Reduces healthcare costs;
3. Saving economic costs of malaria, as workforce is protected;
4. AVP

Against

[1]

1. Not totally effective as only 56% effective / 8.5% still contract malaria;
2. Unknown effects on people older than 17 months / limited ethnic mix in terms of test subjects and therefore unknown effects on other ethnic groups;
3. Small sample size for trial / only one trial conducted;
4. New strains of malaria might appear and being vaccinated with RTS,S will not protect from infection by the new strains;
5. No information on whether there are side effects with the RTS,S vaccine;
6. May be expensive to produce / store / administer the RTS,S vaccine on a large-scale;
7. AVP

[Total: 6]

- 11** Tetanus is a very serious disease and can be fatal. Symptoms of the disease are caused by a toxin produced by *C. tetani*.

There are different ways to gain immunity to the disease, such as described in situations A and B shown in Table 11.1.

Table 11.1

situation	description of immunity
A	A student has immunity after receiving three rounds of a vaccine, which consists of modified tetanus toxin.
B	An unvaccinated person, who has just been infected by the pathogen, gains immediate immunity by being given an injection containing antibodies specific to the tetanus toxin.

- (a)** Compare the type of immunity described in situations A and B. [2]

1. In both A and B, the immunity is artificially acquired;
2. The immunity described in situation A is active while that of situation B is passive;

Points 1+2 cannot be combined– max 1m for A is artificial active vs B is artificial passive

- (b)** *C. tetani* infection can be treated with antibiotics.

Explain if penicillin, which is an antibiotic, will be effective in the treatment of infections caused by other pathogens, such as viruses. [2]

1. Penicillin will be ineffective in treating viral infections;
2. This is because penicillin prevents the formation of peptidoglycan cell wall which is absent in viruses;

Allergens are molecules that can cause allergic diseases. Allergens present on pollen produced by some species of *Bromus* can result in hayfever (allergic rhinitis) in some people.

Immunoglobulin E (IgE) is the class of antibody produced in response to allergens. IgE has a high affinity for Fc receptors on mast cells, which are a type of white blood cell.

Fig. 11.1 is an outline summary of how hayfever develops after an initial 'sensitisation' process.

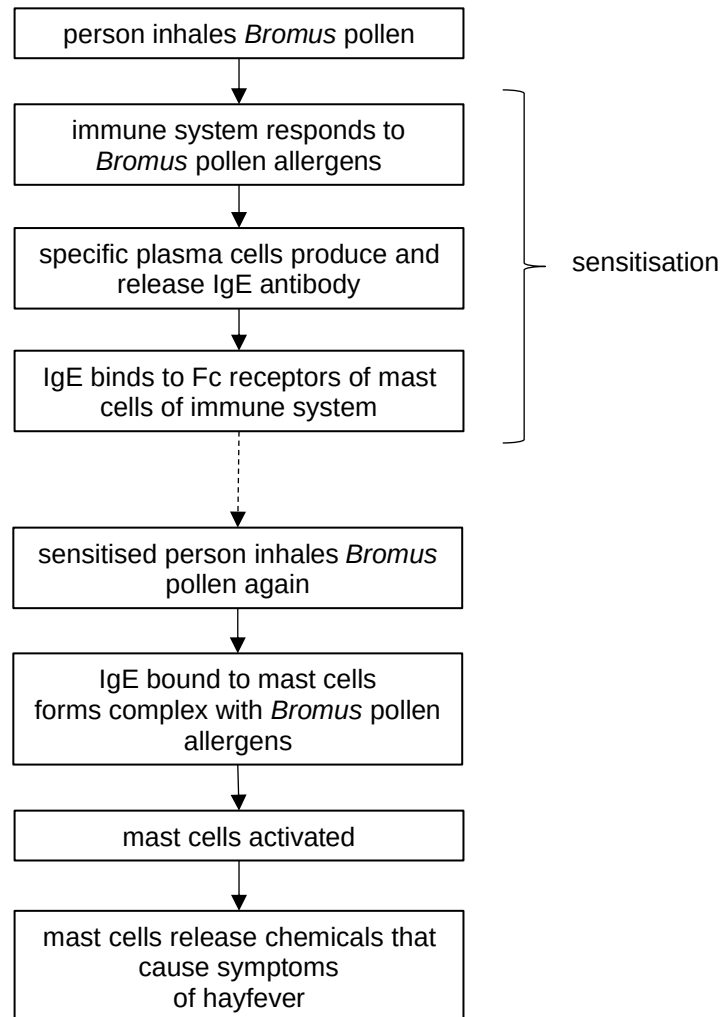


Fig. 11.1

- (c) Omalizumab is an antibody that binds to IgE molecules, and can be used in the treatment of hayfever.

Suggest how Omalizumab may act to reduce the symptoms of hayfever in a sensitised person. [2]

1. Omalizumab binds to the antigen-binding site of IgE and blocks binding of pollen allergens; (R: bind to Fc region of IgE)
2. Mast cells are not activated and will not release chemicals that cause hayfever symptoms;

[Total: 6]